

Branch Specialization in Attention

Toy Head-Dimension Intervention Checkpoint

Research checkpoint

May 21, 2026

RQ5: Does structural heterogeneity cause more stable functional specialization?

This checkpoint tests the intervention in a controlled toy setting before moving to larger language models.

Task:

$$[k_1, v_1, \dots, k_8, v_8, k_q] \rightarrow v_q$$

Experimental Arms

All models use $d_{model} = 128$, two layers, five seeds, and total attention head dimension 128.

Config	Head dimensions	Purpose
uniform4	[32,32,32,32]	same-head-count baseline
hetero4	[16,16,32,64]	heterogeneous intervention
uniform2	[64,64]	fewer/wider control
64-first	[64,16,16,32]	position control

Primary toy specialization score:

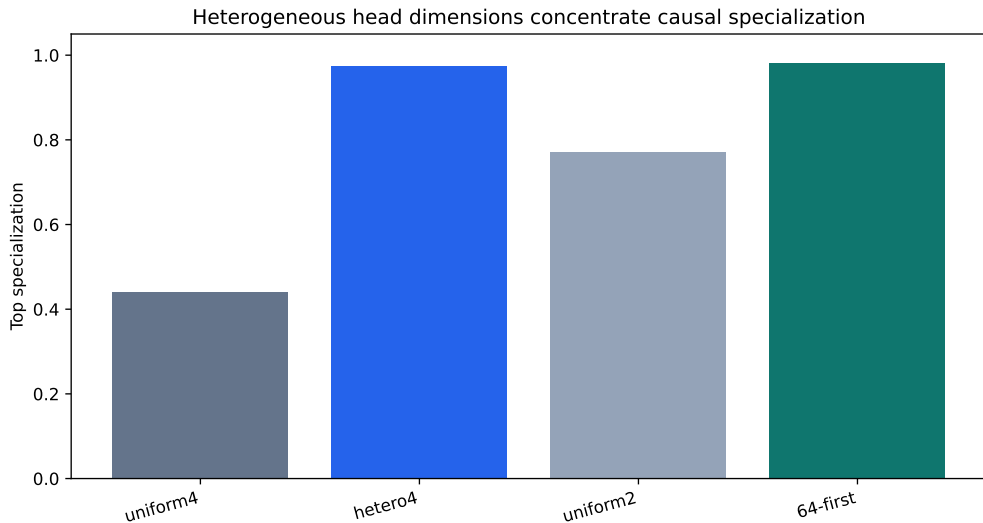
$$S(h, t) = \max(\text{loss}_{\text{ablate}(h)} - \text{loss}_{\text{base}}, 0)$$

Scores are normalized across heads within a layer. This is causal and avoids relying on attention mass alone.

Config	Eval accuracy	Eval loss
uniform4	0.9992	0.0057
hetero4	0.9980	0.0117
uniform2	0.9996	0.0051
64-first	1.0000	0.0032

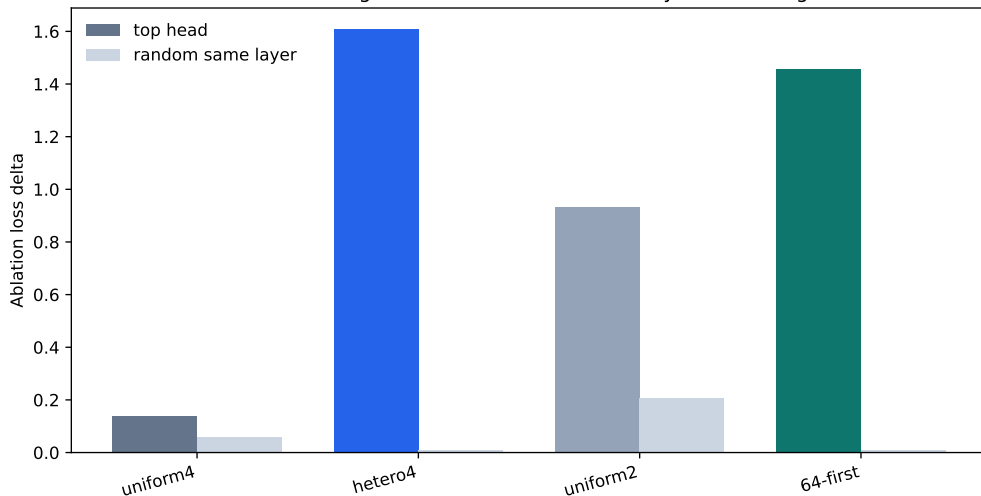
The specialization comparison is not driven by task failure.

Specialization Concentration

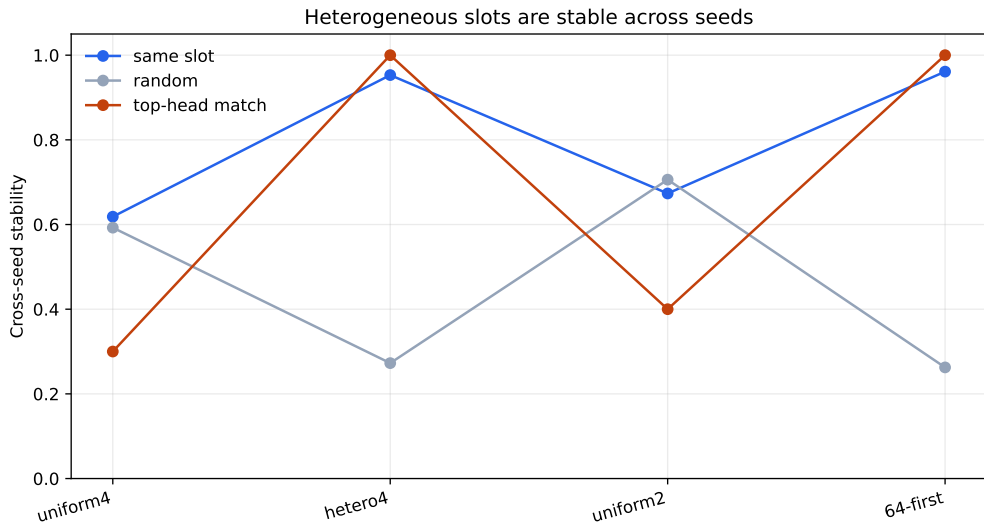


Causal Load

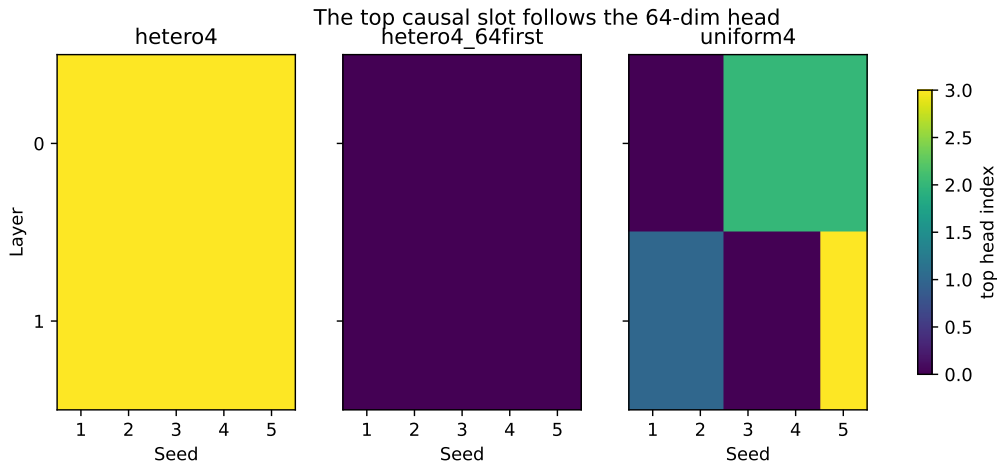
The heterogeneous 64-dim slot is causally load-bearing



Cross-Seed Stability



Position Control



Findings:

- Heterogeneous head dimensions produce a near-single-head causal solution.
- The top causal role follows the 64-dim head when its index changes.
- Uniform fewer/wider heads help, but do not give the same stable slot.
- In heterogeneous architectures, same-slot matching can be more meaningful than unconstrained Hungarian matching.

Continue the architectural-intervention branch.

Next decisive experiment:

- scale this to a richer toy or small LM objective;
- keep uniform same-head-count, heterogeneous, and uniform fewer/wider controls;
- report same-slot and Hungarian-aligned stability side by side.

- Script: `scripts/toy_head_dim_intervention.py`
- Main output: `results/phase3_toy_head_dim_intervention`
- Position control: `results/phase3_toy_head_dim_position_control`
- Report: `doc/phase3_toy_head_dim_intervention.md`
- Data copied into this deck folder under `data/`